Customer Activity & Trends Analysis Insights

Prepared by: Shoaib Alam

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Tools Used:

Python (Pandas, NumPy, Matplotlib, Seaborn)

SQL (MySQL)

Power BI

Jupyter Notebook

1. Project Overview

This project analyzes customer shopping behavior to uncover insights related to revenue generation, customer segmentation, seasonal trends, and product performance.

The objective of this project is to transform raw customer transaction data into meaningful business insights that can support strategic decision-making in retail businesses.

The project follows a complete end-to-end data analytics workflow including:

- **Data Cleaning & Preprocessing** – Python (Pandas)
- **Data Storage & Querying** – SQL (MySQL)
- **Exploratory Data Analysis** – Python (Matplotlib, Seaborn)
- **Dashboard Development** – Power BI

2. Dataset Description

- Rows: 3,900
- Columns: 18
- Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)

- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

3. Data Pipeline (Python Automation)

- Automated logging for error tracking
- Configurable file paths and database connections
- Modular functions for reusability
- Error handling at each stage

The pipeline automates the data preparation process including loading, cleaning, transformation, and storing the processed dataset in a MySQL database.

```
2026-03-09 11:38:23,754 | INFO | ========================================
2026-03-09 11:38:23,754 | INFO | DATA PIPELINE STARTED
2026-03-09 11:38:23,754 | INFO | ========================================
2026-03-09 11:38:23,754 | INFO | Loading Raw Data
2026-03-09 11:38:23,767 | INFO | Data loaded successfully | Shape: (3900, 18)
2026-03-09 11:38:23,767 | INFO | ========================================
2026-03-09 11:38:23,767 | INFO | Initial Data Preview
2026-03-09 11:38:23,767 | INFO | ========================================
2026-03-09 11:38:23,777 | INFO | Initial Preview:    Customer ID  Age Gender Item Purchased  ... Promo Code Used  Previous Purchases Payment Method Frequenc
Purchases
0           1   55   Male        Blouse  ...            Yes                 14         Venmo           Fortnightly
1           2   19   Male       Sweater  ...            Yes                  2          Cash           Fortnightly
2           3   50   Male         Jeans  ...            Yes                 23   Credit Card                Weekly
3           4   21   Male       Sandals  ...            Yes                 49        PayPal                Weekly
4           5   45   Male        Blouse  ...            Yes                 31        PayPal              Annually
5           6   46   Male      Sneakers  ...            Yes                 14         Venmo                Weekly
6           7   63   Male         Shirt  ...            Yes                 49          Cash             Quarterly
7           8   27   Male        Shorts  ...            Yes                 19   Credit Card                Weekly
8           9   26   Male          Coat  ...            Yes                  8         Venmo              Annually
9          10   57   Male       Handbag  ...            Yes                  4          Cash             Quarterly
```

4. DATA CLEANING & PREPROCESSING

Data Quality Assessment

Issue	Findings	Action Taken
Missing Values	Review Rating (37 missing)	Category-wise median imputation

Duplicates	0 duplicates	No action needed
Column Names	Spaces, special characters	Standardized to lowercase with underscores

```
df.isna().sum()
```

```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating            37
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

Handling Missing Values

```
[10]: df['review_rating'] = df.groupby('category')['review_rating'].transform(lambda x: x.fillna(x.median()))
```

```
[11]: df.isna().sum()
```

```
[11]: customer_id               0
      age                       0
      gender                    0
      item_purchased            0
      category                  0
      purchase_amount           0
      location                  0
      size                      0
      color                     0
      season                    0
      review_rating             0
      subscription_status       0
      shipping_type             0
      discount_applied          0
      promo_code_used           0
      previous_purchases        0
      payment_method            0
      frequency_of_purchases    0
      dtype: int64
```

Column Name Standardization

```
•[23]: #Before Standardization Columns
       df.columns
```

```
[23]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
             'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
             'review_rating', 'subscription_status', 'shipping_type',
             'discount_applied', 'promo_code_used', 'previous_purchases',
             'payment_method', 'frequency_of_purchases'],
            dtype='object')
```

```
[22]: df.columns = (
          df.columns
              .str.strip()
              .str.replace(' ', '_')
              .str.lower()
      )
```

```
•[24]: #After Standardization Columns
       df = df.rename(columns={'purchase_amount_(usd)':'purchase_amount'})
       df.columns
```

```
[24]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
             'purchase_amount', 'location', 'size', 'color', 'season',
             'review_rating', 'subscription_status', 'shipping_type',
             'discount_applied', 'promo_code_used', 'previous_purchases',
             'payment_method', 'frequency_of_purchases'],
            dtype='object')
```

5. Feature Engineering

Age Group Segmentation

Feature engineering was performed to create additional analytical variables that help understand customer behavior more effectively.

```
[12]: df['age_group'] = pd.cut(
          df['age'],
          bins=[0, 25, 40, 60, 100],
          labels=['Young Adult', 'Young', 'Middle Aged', 'Senior']
      )
      df.columns
```

```
[12]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
             'purchase_amount', 'location', 'size', 'color', 'season',
             'review_rating', 'subscription_status', 'shipping_type',
             'discount_applied', 'promo_code_used', 'previous_purchases',
             'payment_method', 'frequency_of_purchases', 'age_group'],
            dtype='object')
```

Customer Segmentation

```
[13]: df.loc[df['previous_purchases'] == 1,  'customer_segmentation'] = 'New'
df.loc[(df['previous_purchases'] >= 2) & (df['previous_purchases'] <= 10), 'customer_segmentation'] = 'Returning'
df.loc[df['previous_purchases'] > 10, 'customer_segmentation'] = 'Loyal'
customer_segmentation = df['customer_segmentation'].value_counts()
customer_segmentation
```

```
[13]: customer_segmentation
Loyal        3116
Returning     701
New            83
Name: count, dtype: int64
```

6. Exploratory Data Analysis (EDA)

We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- The dataset structure was examined using **df.info()** and summary statistics were generated using **df.describe()**.

```
df.describe(include='all')
```

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3900.000000	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750051	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.713590	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN

shipping_type	discount_applied	promo_code_used	previous_purchases	payment_method	frequency_of_purchases	age_group	customer_segmentation
3900	3900	3900	3900.000000	3900	3900	3900	3900
6	2	2	NaN	6	7	4	3
Free Shipping	No	No	NaN	PayPal	Every 3 Months	Middle Aged	Loyal
675	2223	2223	NaN	677	584	1503	3116
NaN	NaN	NaN	25.351538	NaN	NaN	NaN	NaN
NaN	NaN	NaN	14.447125	NaN	NaN	NaN	NaN
NaN	NaN	NaN	1.000000	NaN	NaN	NaN	NaN
NaN	NaN	NaN	13.000000	NaN	NaN	NaN	NaN
NaN	NaN	NaN	25.000000	NaN	NaN	NaN	NaN
NaN	NaN	NaN	38.000000	NaN	NaN	NaN	NaN
NaN	NaN	NaN	50.000000	NaN	NaN	NaN	NaN

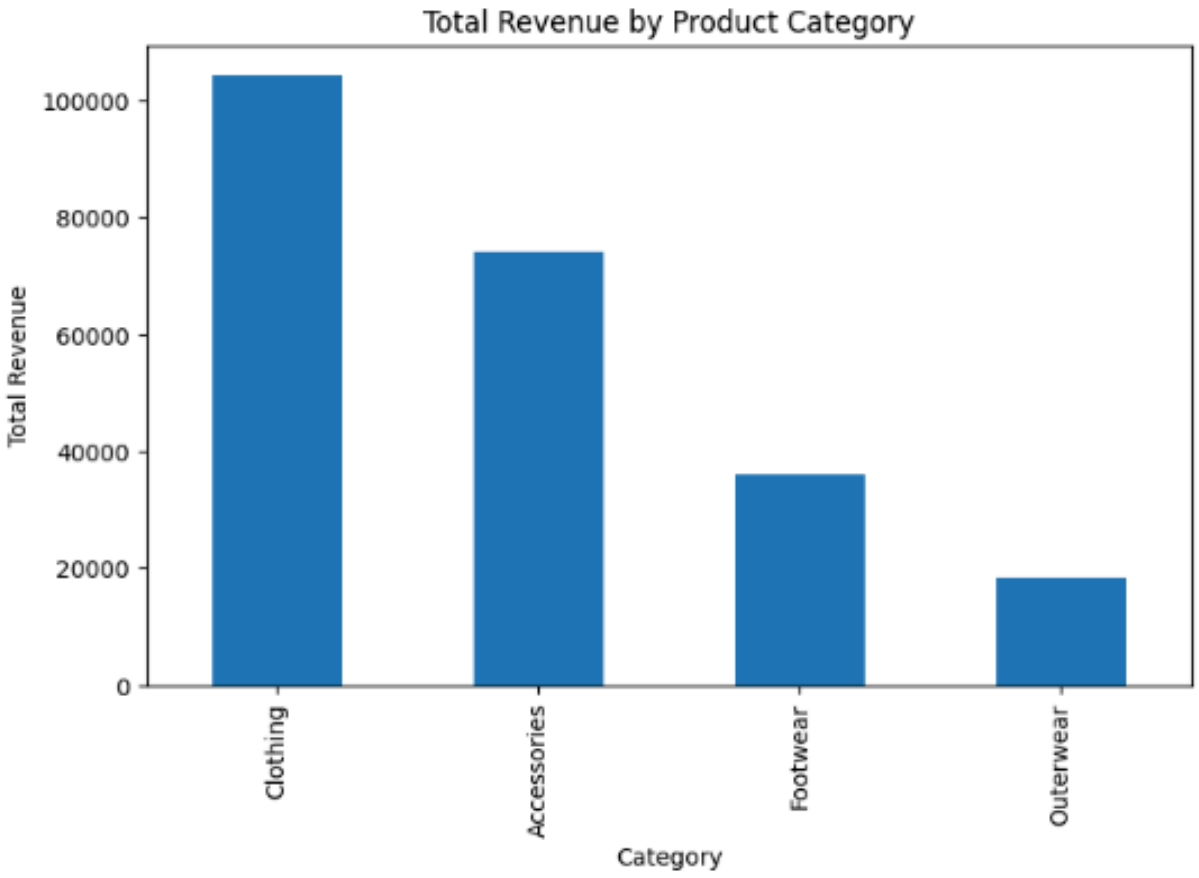
- **Missing Data Handling**: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization**: Renamed columns to snake case for better readability and documentation.
- **Feature Engineering**:
  - Created `age_group` column by binning customer ages.
  - Created `customer_segmentation` column from purchase data.
- **Data Consistency Check**: Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration**: Connected Python script to My SQL and loaded the cleaned DataFrame into the database for SQL analysis.

Revenue Distribution by Category

[19]:
```
# Total Revenue by Category
category_revenue = (
    df.groupby('category')['purchase_amount']
        .sum()
        .sort_values(ascending=False)
)
category_revenue
```

[19]:
```
category
Clothing       104264
Accessories     74200
Footwear        36093
Outerwear       18524
Name: purchase_amount, dtype: int64
```

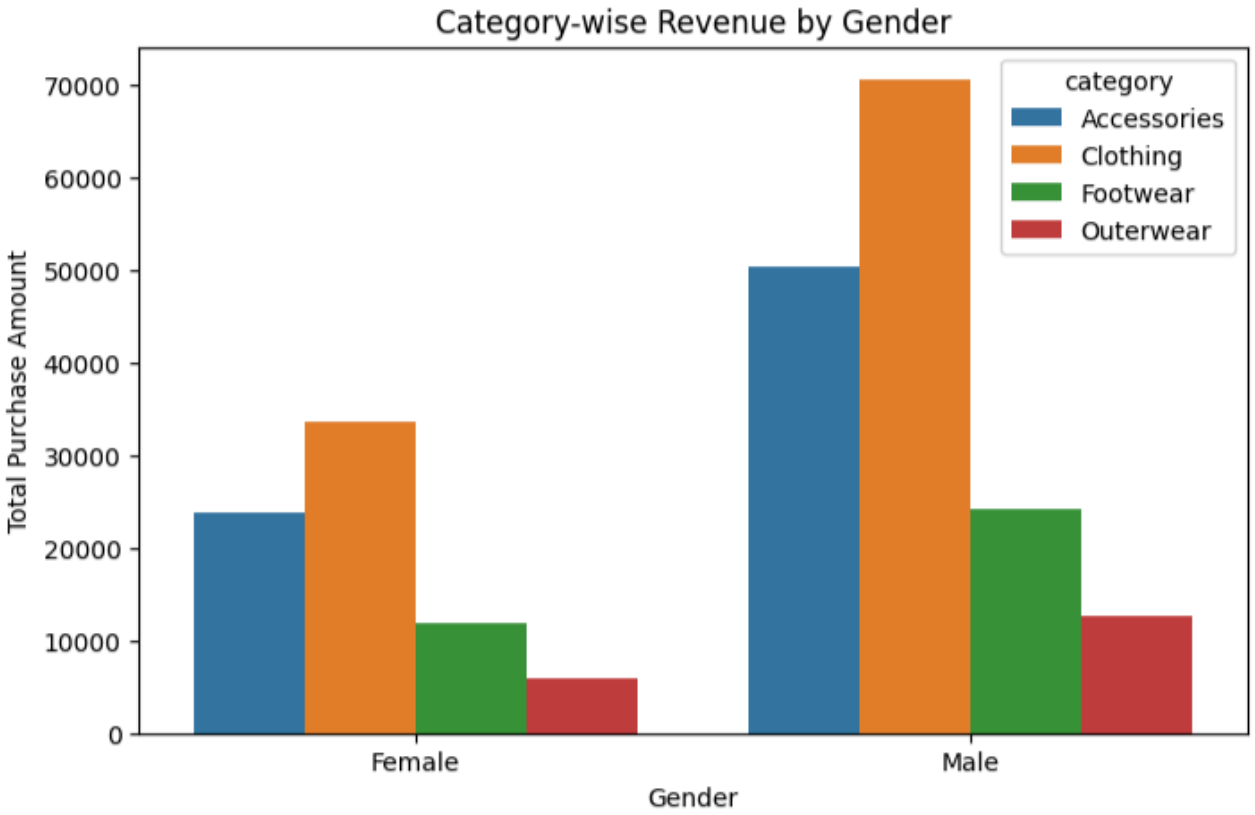
[20]:
```
plt.figure(figsize=(8, 5))
category_revenue.plot(kind='bar')
plt.title('Total Revenue by Product Category')
plt.xlabel('Category')
plt.ylabel('Total Revenue')
plt.show()
```



Revenue Distribution by Gender

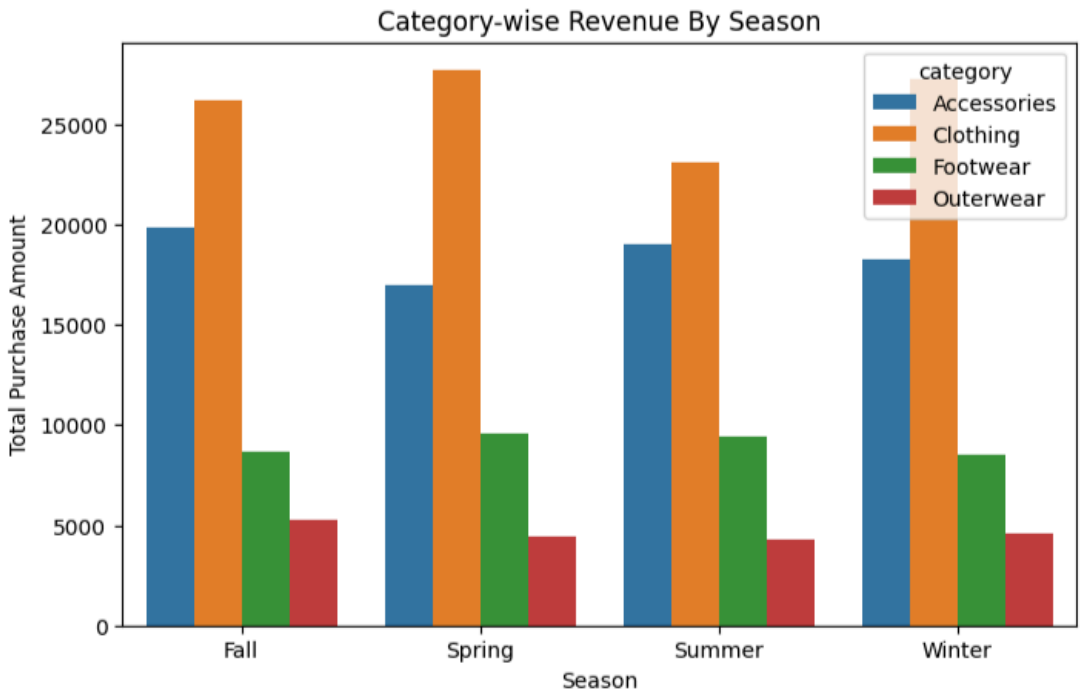
[28]:
```
plt.figure(figsize=(8,5))
sns.barplot(
    data=revenue_by_gender_category,
    x='gender',
    y='purchase_amount',
    hue='category'
)

plt.title('Category-wise Revenue by Gender')
plt.xlabel('Gender')
plt.ylabel('Total Purchase Amount')
plt.show()
```



Seasonal Revenue Trends

```
[30]: plt.figure(figsize=(8,5))
sns.barplot(
    data=revenue_by_season_category,
    x='season',
    y='purchase_amount',
    hue='category',
)
plt.title('Category-wise Revenue By Season')
plt.xlabel('Season')
plt.ylabel('Total Purchase Amount')
plt.show()
```



7. SQL Business Analysis

We performed structured analysis in MySQL to answer key business question:

1. **Revenue by Gender-** Compared total revenue generated by male vs female customer.

[10]:

	gender	customer_count
0	Female	1248
1	Male	2652

2. **High-Spending Discount Users -** Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	total_revenue
0	2	64.0
1	3	73.0
2	4	90.0
3	7	85.0
4	9	97.0
...	...	...
834	1667	64.0
835	1671	73.0
836	1673	73.0
837	1674	62.0
838	1676	90.0

839 rows × 2 columns

3. **Top 5 Products by Rating -** Found products with the highest average review rating.

	item_purchased	avg_rating
0	Gloves	3.86
1	Sandals	3.84
2	Boots	3.82
3	Hat	3.80
4	Handbag	3.78

4. **Shipping Type Comparison -** Compared average purchase amounts between Standard and Express shipping.

	shipping_type	avg_purchase_amount
0	Express	60.48
1	Standard	58.55

5. **Subscribers vs Non-Subscribers -** Compared average spend and total revenue across subscription status.

	subscription_status	total_revenue	avg_revenue
0	No	170436.0	59.8651
1	Yes	62701.0	59.5451

6. **Discount-Dependent Product -** Identified 5 products with the highest percentage of discounted purchases.

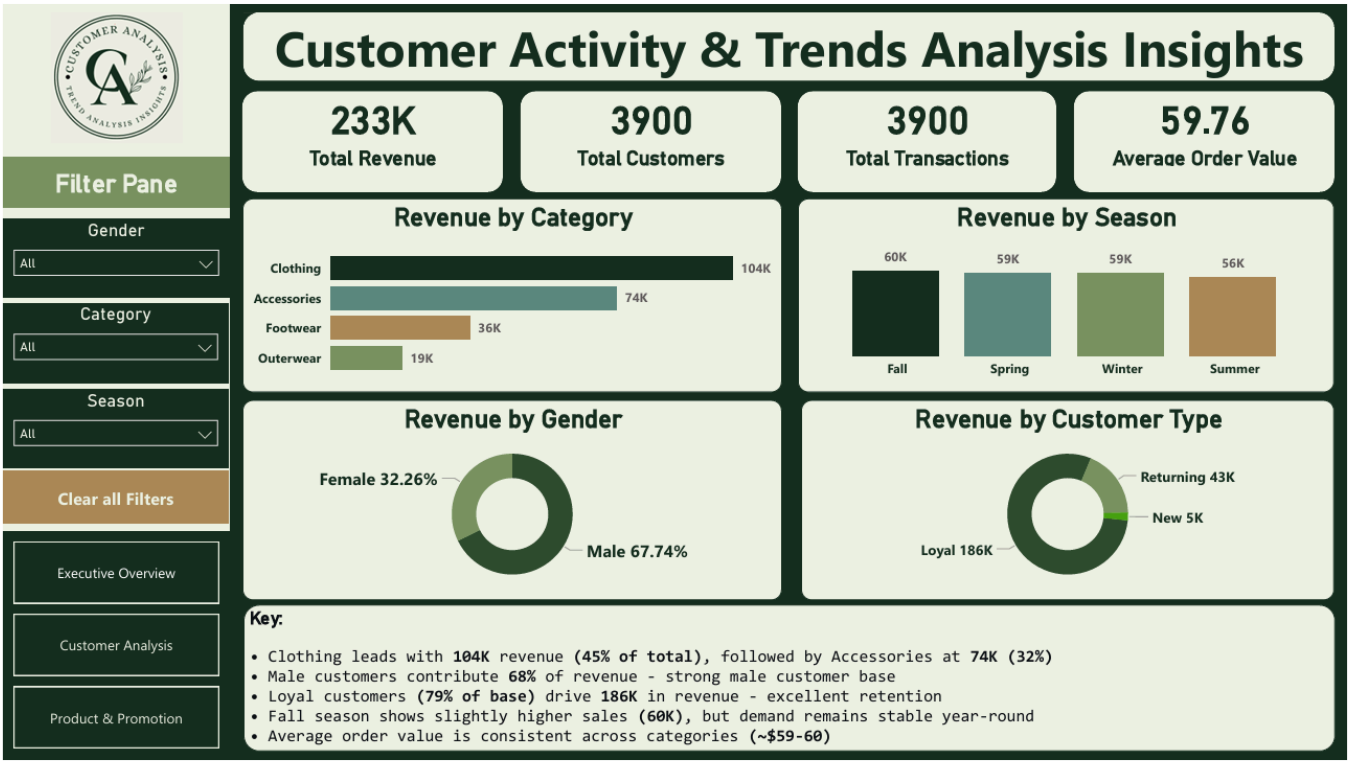
	item_purchased	discount_percentage
0	Sneakers	50.05
1	Hat	49.02
2	Sweater	48.72
3	Pants	48.23
4	Coat	48.19

7. **Customer Segmentation -** Classified customers into New, Returning , and Loyal segments based on purchase history.

	customer_segmentation	number_of_customer
0	Loyal	3116
1	New	83
2	Returning	701

8. Power BI Dashboard

Executive Overview Dashboard



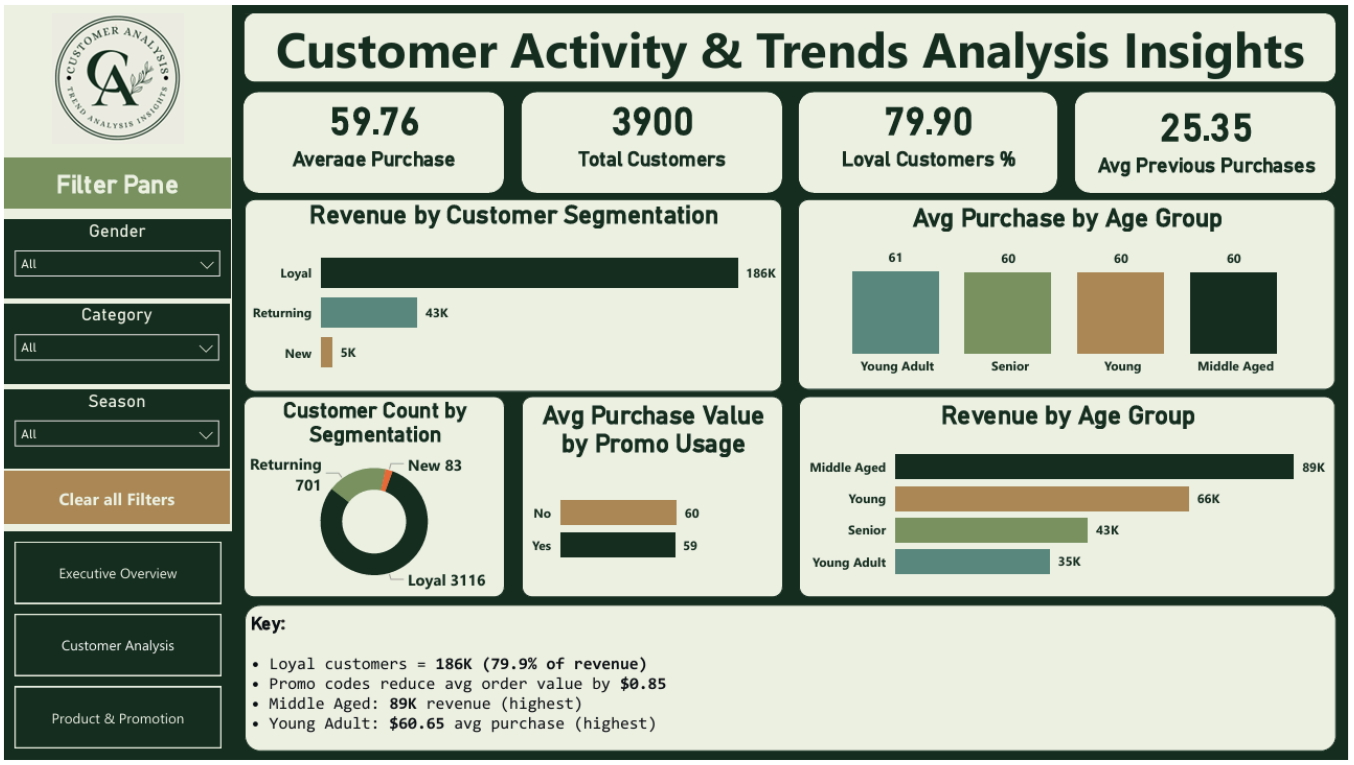
Key Insights:

- Total Revenue: **233K**
- Total Customers: **3900**
- Average Order Value: **59.76**

Clothing contributes the largest revenue share at **104K**, followed by Accessories at **74K**.

Male customers contribute **68% of total revenue**.

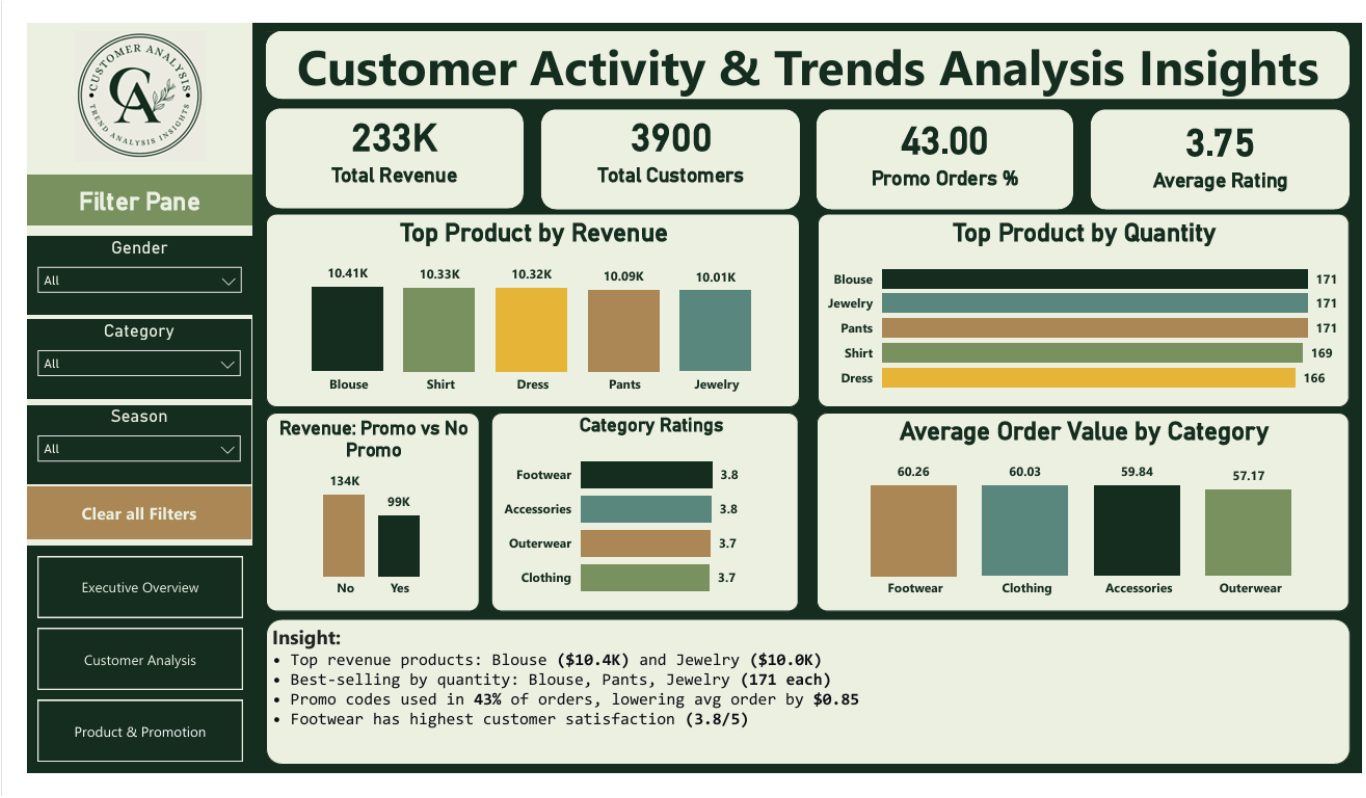
Customer Analysis Dashboard



Key Insights:

- Loyal customers generate the majority of revenue, contributing **186K**, which indicates strong customer retention.
- Middle-aged customers generate the highest revenue contribution among age groups.

Product & Promotion Analysis



Key Insights:

- Blouse, Shirt, and Dress are among the top revenue-generating products.
- Promo codes are used in 43% of orders, but they slightly reduce average order value.
- Footwear has the highest customer satisfaction rating.

9. Key Business Insights

• Loyal customers generate the majority of revenue
• Clothing category dominates sales performance
• Promo usage impacts order value slightly
• Middle-aged customers represent the most valuable segment
• Product demand is relatively stable across seasons

10. Business Recommendations

1. Focus marketing campaigns on loyal customers to maintain retention.

2. Expand product offerings in the clothing category.

3. Optimize promotional strategies to improve order value.

4. Target middle-aged customers through personalized marketing.

11. Conclusion

This project demonstrates the complete end-to-end data analytics process, from raw data preparation to business insight generation.

By combining Python, SQL, and Power BI, the project transforms raw transactional data into actionable insights that can support data-driven decision-making in retail businesses.